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4.4 Atomic Structure

AQA GCSE Physics (8463) — Paper 1

4.4.1 The Structure of the Atom

Atoms are the smallest indivisible units of matter and consist of a nucleus (containing protons and neutrons) surrounded by electrons in electron shells. Atoms are electrically neutral overall —the positive charge of the nucleus is balanced by the negative charge of the electrons.

Every atom has three main types of subatomic particles:

- **Protons** — located in the nucleus, charge +1 (relative), mass ≈ 1 u (atomic mass unit)
- **Neutrons** — located in the nucleus, charge 0 (neutral), mass ≈ 1 u
- **Electrons** — orbiting the nucleus in shells, charge −1 (relative), mass ≈ 1/1836 u (negligible)

Size of the Atom and Nucleus

Nuclear radius: ~10^{-15} m (1 femtometre, fm).
Atomic radius: ~10^{-10} m (1 ångström, Å).
The nucleus occupies only about 1/10 000 of the atom's radius, yet contains almost all the atom's mass.

To visualize this scale: if the nucleus were the size of a marble (1 cm diameter), the atom would be a sphere about 100 metres in diameter. The atom is mostly empty space.

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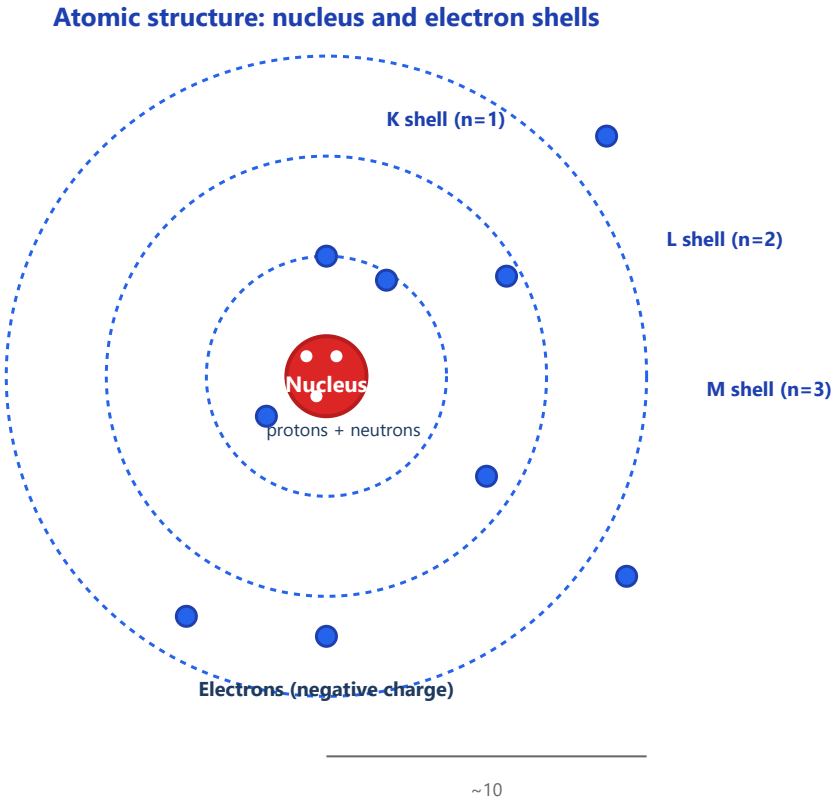

$^{-10}$ m (atomic radius) ~10^{-15} m

Electrical Charge and Neutrality

> Each **proton** carries a charge of +1 (in relative units). Each **electron** carries a charge of −1. **Neutrons are electrically neutral** (charge = 0). In a neutral atom, the number of protons equals the number of electrons, so the total charge is zero.

Ions: An atom becomes an ion when it loses or gains electrons:

- **Positive ion (cation):** the atom has lost one or more electrons. Example: Na^+ has 11 protons but only 10 electrons.
- **Negative ion (anion):** the atom has gained one or more electrons. Example: Cl^- has 17 protons but 18 electrons.

> **Exam Tip:** When counting charge, remember: protons are in the nucleus and don't change during ionisation. Only electrons are added or removed. If an atom loses electrons, it becomes

positively charged (cation). If it gains electrons, it becomes negatively charged (anion).

4.4.2 Mass Number, Atomic Number and Isotopes

Atomic number (Z): the number of protons in the nucleus. This number defines which element the atom is. Every atom of carbon has 6 protons (Z = 6); every atom of oxygen has 8 protons (Z = 8).

Mass number (A): the total number of protons plus neutrons in the nucleus. Since protons and neutrons each have a mass of approximately 1 u, the mass number is approximately the relative atomic mass of the atom.

Notation: An atom is represented as ^{A}ZX, where X is the chemical symbol.

- Example: $^{14}{}_{6}C$ (Carbon-14) has A = 14, Z = 6. It contains 6 protons and 14 − 6 = 8 neutrons.
- Example: $^{16}{}_{8}O$ (Oxygen-16) has A = 16, Z = 8. It contains 8 protons and 16 − 8 = 8 neutrons.

Isotopes

Isotopes are atoms of the same element (same Z, same number of protons) that have different numbers of neutrons (different A, different mass numbers). They have the same chemical properties but different physical properties (especially mass and nuclear stability).

Example: Carbon has two stable isotopes:

- **Carbon-12:** $^{12}{}_{6}C$ — 6 protons, 6 neutrons, mass ≈ 12 u (used as the standard for atomic mass)
- **Carbon-14:** $^{14}{}_{6}C$ — 6 protons, 8 neutrons, mass ≈ 14 u (radioactive; used for radiocarbon dating)

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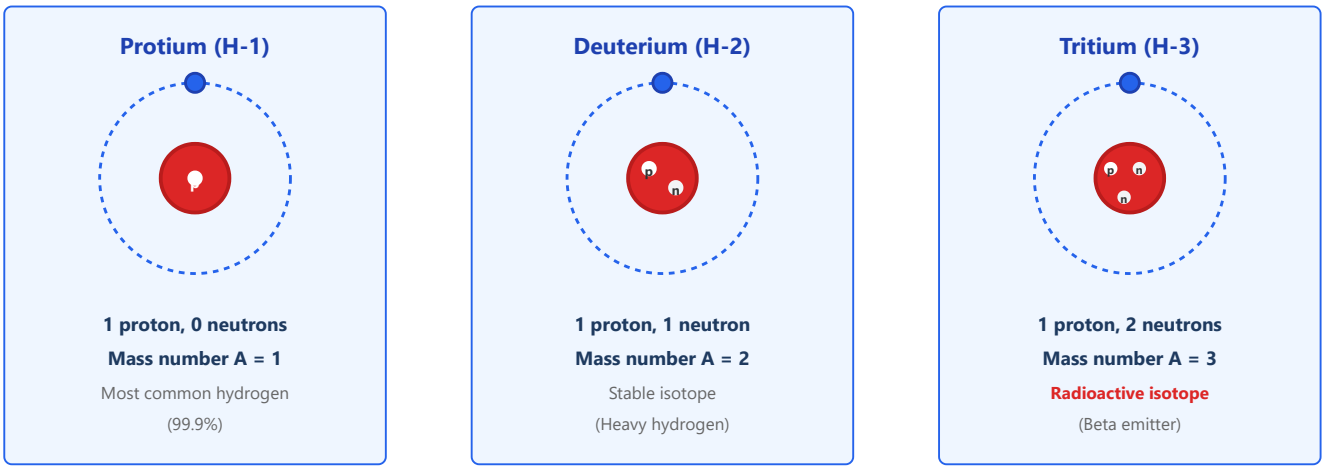
Isotopes of hydrogen: same protons, different neutrons



4.4.3 Development of the Model of the Atom (Historical)

Our understanding of atomic structure has evolved through centuries of experimental evidence and theoretical progress:

Ancient Greeks (~440 BCE)

Democritus proposed that all matter consists of tiny, indivisible particles called atoms. This was pure philosophy—no experimental evidence.

Dalton's Atomic Theory (~1808)

John Dalton revived atomism and proposed that atoms are solid, indivisible spheres. Different elements have atoms of different masses. His theory explained the law of conservation of mass and the law of definite proportions in chemical reactions.

Thomson's "Plum Pudding" Model (1897)

J.J. Thomson discovered the electron using cathode ray tubes. He showed that atoms are not indivisible but contain negatively charged particles. His "plum pudding" model pictured a positively charged sphere with electrons embedded in it like plums in pudding.

Rutherford's Nuclear Model (1909)

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Ernest Rutherford conducted the gold-foil scattering experiment. He bombarded thin gold foil with alpha particles (helium nuclei) and observed their deflection:

- Most alpha particles passed straight through the foil
- Some particles were deflected at large angles
- A few particles bounced backward

This proved that the atom is mostly empty space with a small, dense, positively charged nucleus at the centre.

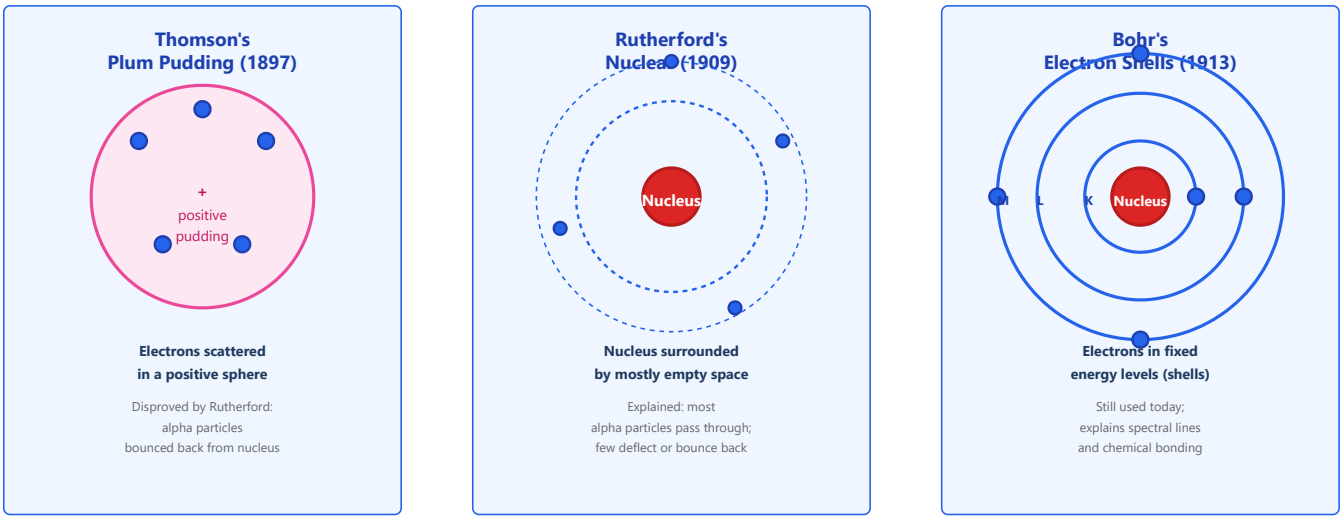
Bohr's Model (1913)

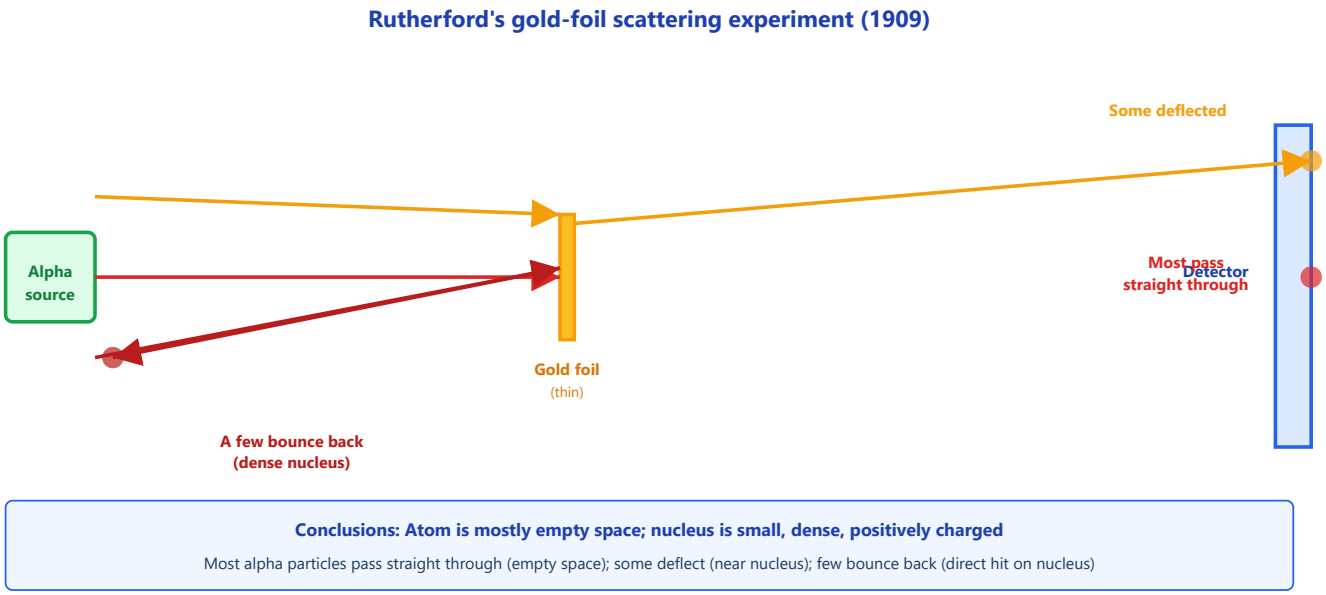
Niels Bohr proposed that electrons orbit the nucleus in fixed shells or energy levels, each with a specific energy. Electrons can jump between shells by absorbing or emitting electromagnetic radiation. This model explained the discrete spectral lines of hydrogen.

Chadwick's Neutron (1932)

James Chadwick discovered the neutron—a neutral particle in the nucleus. This completed the basic picture: atoms consist of protons and neutrons in a nucleus, surrounded by electrons in shells.

Evolution of atomic models





Exam Tip: Examiners often ask why Rutherford's results disproved the plum pudding model. The answer is: if electrons were embedded in a positive pudding, alpha particles would be deflected gradually as they passed through. Instead, most passed straight (empty space) and some deflected sharply (dense nucleus). This proved a concentrated nucleus, not a diffuse positive sphere.

4.4.4 Electron Energy Levels

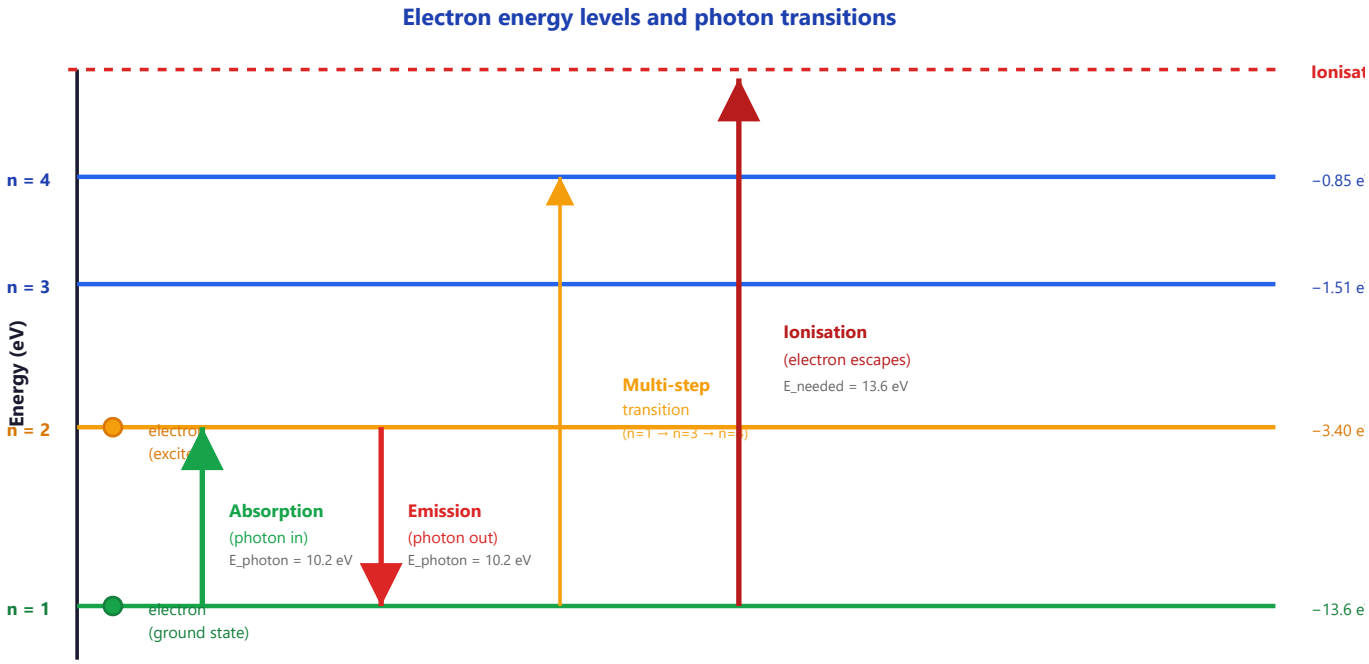
Electrons in an atom occupy **discrete energy levels** (or shells, K, L, M, N, etc.). An electron in a lower level has less energy; in a higher level, more energy. Electrons cannot exist between levels—they can only jump between specific energy levels by absorbing or emitting precise amounts of energy.

Excitation and Ionisation

Excitation: When an electron absorbs energy (e.g., from an electromagnetic photon), it jumps to a higher energy level. The atom is now in an "excited state."

De-excitation: An excited electron quickly drops back to a lower level, emitting energy as a photon. The energy of the photon equals the energy difference between the two levels.

Ionisation: If an electron absorbs enough energy, it can escape the atom entirely (goes to energy level infinity). The atom becomes positively charged (cation). The minimum energy needed is called the ionisation energy.



Exam Tip: A key exam concept: the energy of an emitted photon equals the energy difference between two levels. If an electron drops from level 3 to level 1, the photon energy is $E_3 - E_1$. This is why different atoms emit different colours of light—each element's energy levels are unique.

4.4.5 Radioactive Decay and Nuclear Radiation

Radioactivity is the spontaneous, random decay of unstable (radioactive) atomic nuclei. The decay process is random—you cannot predict which atom will decay or when. Radioactive nuclei emit one or more types of nuclear radiation.

Types of Nuclear Radiation

Alpha (α) particles: Helium-4 nuclei (^{2}He or $^{4}_{2}$He). Composition: 2 protons + 2 neutrons. Charge: +2 (relative). Mass: ~4 u.

Penetration: Stopped by a few centimetres of air or thin paper.
Ionisation: Highly ionising (strips electrons from many atoms).

Beta-minus (β^-) particles: Fast-moving electrons emitted when a neutron converts to a proton (and an antineutrino). Charge: −1. Mass: ~0 (1/1836 u).

Penetration: Stopped by a few millimetres of aluminium.
Ionisation: Moderately ionising.

Gamma (γ) rays: High-energy electromagnetic radiation (photons) emitted from the nucleus. No mass, no charge.

Penetration: Very penetrating; needs several centimetres of lead or metres of concrete.
Ionisation: Weakly ionising (but very penetrating, so still dangerous).

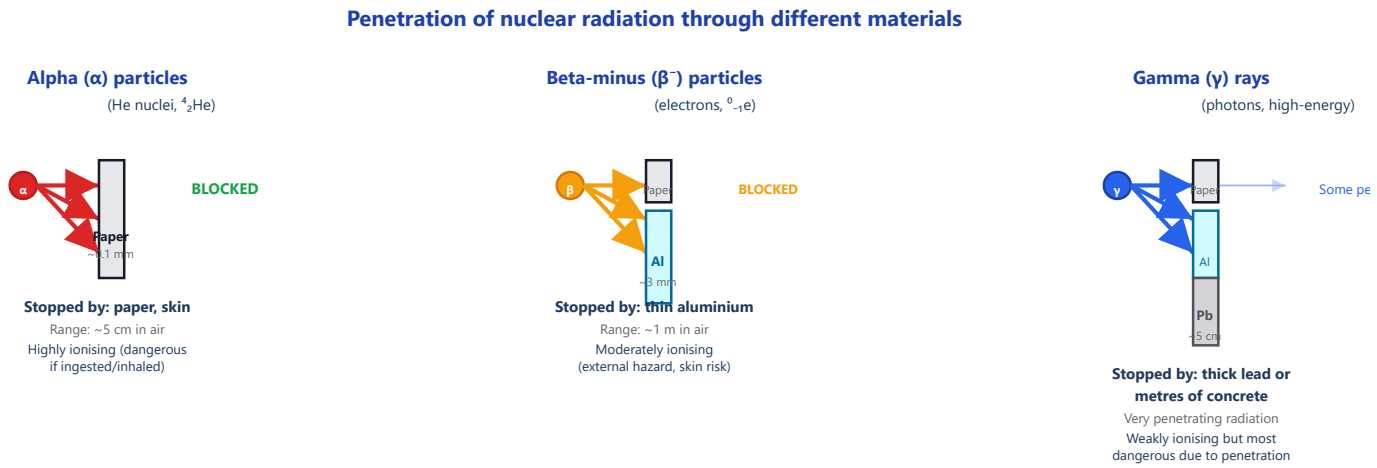


Table: Properties of Nuclear Radiation

Radiation	Composition	Mass (u)	Charge	Ionisation	Penetration	Deflection by E/M field
Alpha (α)	^{2}He nucleus (2p + 2n)	4	+2	Highly ionising	Stopped by paper or 5 cm air	Deflected toward cathode (−)
Beta-minus (β^-)	Electron ($^{0}_{-1}e$)	~0 (1/1836)	−1	Moderately ionising	Stopped by 3 mm Al	Deflected toward anode (+)
Gamma (γ)	Photon (EM radiation)	0	0	Weakly ionising	Stopped by 5 cm Pb or metres of concrete	No deflection (uncharged)

Exam Tip: A key exam question: "Name and describe three types of nuclear radiation." Always give the name, the composition (what it is), its charge, and its penetrating power. Alpha = helium nucleus, large and slow, stopped by paper. Beta = electron, fast, needs aluminium. Gamma = light/photons, very fast, needs lead.

4.4.6 Nuclear Equations

In radioactive decay, the **mass number A** and **atomic number Z** are conserved (they stay the same before and after). These conservation laws allow us to predict the products of nuclear decay.

Alpha Decay

An alpha particle (helium nucleus) is emitted. The parent nucleus loses 2 protons and 2 neutrons:

```
General form:  ᴬZX → ᴬ⁻⁴Z⁻₂Y + ⁴₂He (or α)
```

```
Example:        ²³⁸₉₂U → ²³⁴₉₀Th + ⁴₂He
                Uranium-238 → Thorium-234 + Alpha particle

Check mass numbers:   238 = 234 + 4 ✓
Check atomic numbers: 92 = 90 + 2 ✓
```

Beta-minus Decay

A neutron converts into a proton and an electron (beta particle). The nucleus gains 1 proton but mass number stays the same:

```
General form:  ᴬZX → ᴬ⁺¹ᶻY + ⁰₋₁e (or β⁻)

Example:        ¹⁴₆C → ¹⁴₇N + ⁰₋₁e
                Carbon-14 → Nitrogen-14 + Beta particle

Check mass numbers:   14 = 14 + 0 ✓
Check atomic numbers: 6 = 7 + (-1) ✓
```

Exam Tip: In beta decay, the mass number doesn't change (the emitted electron has negligible mass, mass ~0). The atomic number increases by 1 (one more proton). This changes the element—Carbon becomes Nitrogen.

Gamma Emission

A nucleus in an excited state emits a high-energy photon and drops to a lower energy state. No particles are emitted; mass and atomic numbers stay the same:

```
General form:  ᴬZX* → ᴬZX + γ
               (excited nucleus) → (nucleus in lower state) + photon

Example:        ⁶⁰₂₇Co* → ⁶⁰₂₇Co + γ
                Cobalt-60 (excited) → Cobalt-60 (ground state) + gamma ray

Check mass numbers:   60 = 60 ✓
Check atomic numbers: 27 = 27 ✓
```

Neutron Emission

In some very heavy nuclei, a free neutron is emitted. Mass number decreases by 1; atomic number stays the same:

```
General form:  ᴬZX → ᴬ⁻¹ZX + ¹₀n

Check mass numbers:   A = (A−1) + 1 ✓
Check atomic numbers: Z = Z + 0 ✓
```

4.4.7 Half-Life and Radioactive Decay

Half-life ($t_{1/2}$): the time taken for half of the radioactive nuclei in a sample to decay (or equivalently, for the activity/count-rate to halve). Different isotopes have vastly different half-lives—from fractions of a second to billions of years.

Activity: the number of decays per second. Symbol: A, unit: becquerel (Bq). 1 Bq = 1 decay per second.

Count-rate: the number of decays detected per second by a detector (e.g., Geiger-Müller tube). It is proportional to activity but depends on detector efficiency.

Half-Life Calculations

After n half-lives, the fraction of nuclei remaining is:

```
Fraction remaining = (1/2)ⁿ = (0.5)ⁿ

After 1 half-life:  1/2 of the original remains
After 2 half-lives: 1/4 remains
After 3 half-lives: 1/8 remains
After 4 half-lives: 1/16 remains
```

Example: Iodine-131 has a half-life of 8 days. A sample has 10 000 Bq of I-131.

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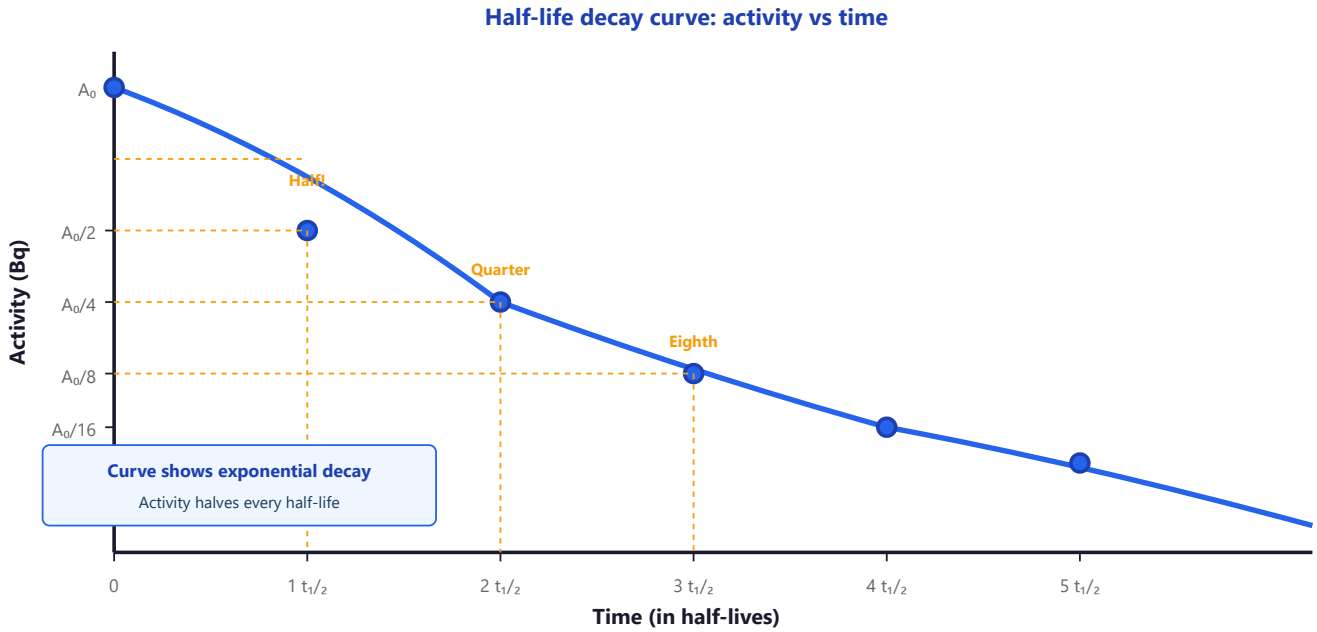
```
After 8 days (1 half-life):     5 000 Bq
After 16 days (2 half-lives):   2 500 Bq
After 24 days (3 half-lives):   1 250 Bq
After 40 days (5 half-lives):   312.5 Bq
```



Half-Life Values for Common Isotopes

Isotope	Half-life	Decay type	Use
Carbon-14	5 730 years	Beta-minus	Radiocarbon dating of archaeological samples
Uranium-235	704 million years	Alpha	Nuclear reactor fuel, bomb material (fissile)
Iodine-131	8.02 days	Beta-minus	Medical tracer, produced in nuclear reactors
Cobalt-60	5.27 years	Beta-minus + gamma	Medical (radiotherapy), sterilisation of instruments

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Polonium-210	138 days	Alpha	Smoke detectors, research

4.4.8 Radioactive Contamination vs Irradiation

Irradiation: being exposed to radiation from an external source. Once you move away from the source, you are no longer irradiated. You yourself are not radioactive.

Example: standing near a radioactive material, or being treated with a gamma beam in a hospital.

Contamination: having radioactive atoms on your body, on surfaces, or inside you. The contaminating atoms continue to decay, emitting radiation into or from you. Contamination is removed by decontamination (washing, removing clothes).

Safety Measures

- **Distance:** Use tongs or remote-handling equipment. Radiation intensity falls with distance squared.
- **Shielding:** Lead-lined boxes for storage. Lead aprons for medical staff. Concrete walls in laboratories.
- **Time:** Minimize exposure time. People working with radioactive materials wear dosimeters to monitor cumulative dose.
- **Protective clothing:** Gloves, lab coats, sometimes full-body suits to prevent contamination.

Uses of Nuclear Radiation

- **Medical imaging:** Gamma-emitting tracers (e.g., Tc-99m) injected into patients. A gamma camera detects emissions and builds an image of internal organs (PET scans, SPECT scans).
- **Cancer treatment (radiotherapy):** Cobalt-60 or linear accelerators deliver targeted gamma or high-energy X-ray beams to tumours, destroying cancer cells.

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- **Smoke detectors:** Contain Americium-241 (alpha emitter). Alpha particles ionise air molecules, creating a small current. Smoke particles block the current, triggering the alarm.
- **Sterilisation:** Cobalt-60 gamma rays kill bacteria and viruses on medical instruments (surgical tools, implants, dressings) without damaging them.
- **Industrial gauges:** Beta or gamma sources measure the thickness of paper, metal sheets, or coatings in manufacturing.

Background Radiation

We are all exposed to a low level of radiation from natural sources:

- **Radon gas (~46%):** Radon-222 seeps from rocks and soil into buildings. This is the largest contributor to background radiation in most regions.
- **Rocks and soil (~18%):** Uranium, thorium, and potassium-40 in granite, limestone, and other minerals.
- **Cosmic rays (~10%):** High-energy particles from space; exposure is higher at higher altitudes and latitudes.
- **Medical procedures (~11%):** X-rays, CT scans, nuclear medicine scans.
- **Nuclear weapons testing (~4%):** (historical, declining).
- **Nuclear accidents and industry (~1%):** (usually very small).

> **Exam Tip:** Examiners often ask about the difference between contamination and irradiation. Key points: Irradiation = exposure to external radiation (stops when source is removed). Contamination = radioactive material on/in you (continues emitting until it decays or is removed). A patient receiving radiotherapy is irradiated. If a worker spills radioactive solution on their skin, they are contaminated.

4.4.9 Nuclear Fission (Triple Science)

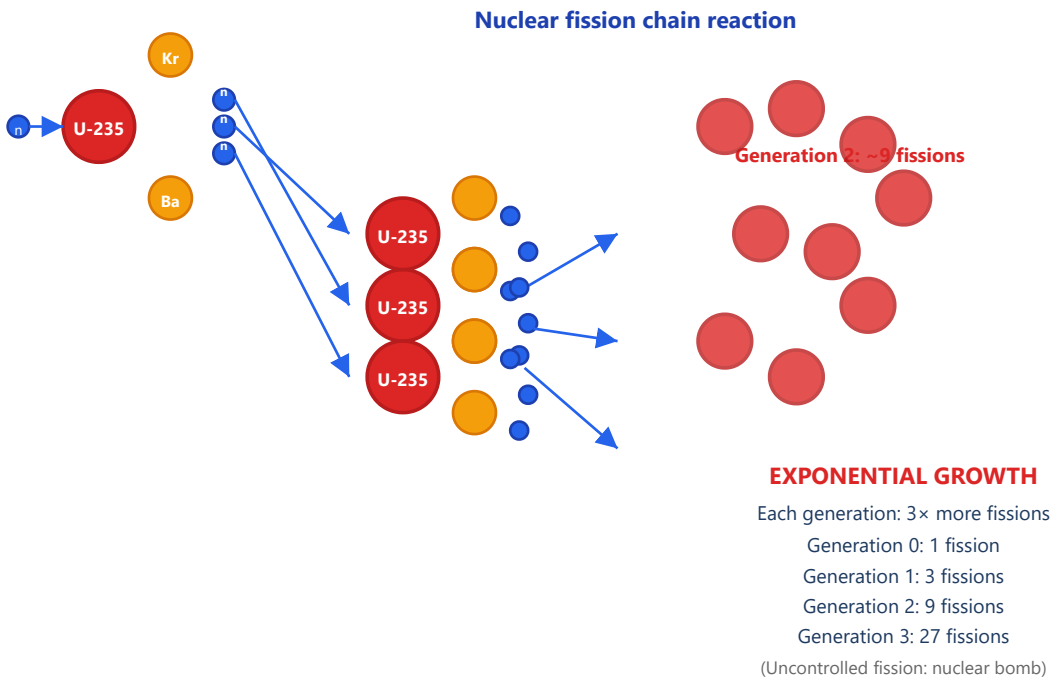
> **Nuclear fission:** the splitting of a large unstable nucleus into two smaller daughter nuclei, releasing energy and neutrons. The process is initiated by the absorption of a neutron (usually

> a slow neutron) and leads to a chain reaction if conditions are right.

The Fission Process

1. **Slow neutron absorbed:** A slow (thermal) neutron collides with a heavy nucleus (e.g., U-235), making it unstable.
2. **Nucleus becomes excited:** The nucleus splits into two daughter nuclei and releases 2–3 additional neutrons plus energy (~200 MeV).
3. **Chain reaction:** The released neutrons can trigger further fissions. If one neutron per fission causes another fission, the reaction is self-sustaining (critical). If more than one neutron causes a fission, the reaction accelerates (supercritical).

```
Typical fission reaction:
²³⁵₉₂U + ¹₀n → ¹⁴⁶₅₆Ba + ⁸⁹₃₆Kr + 3¹₀n + ~200 MeV

Note: Different daughter nuclei are produced in each fission.
Mass number: 235 + 1 = 146 + 89 + 3(1) = 238 ✓
Atomic number: 92 + 0 = 56 + 36 + 0 = 92 ✓
```



Nuclear Reactor Operation

- **Fuel:** Enriched uranium (U-235, typically 3–5% enrichment) or natural uranium in some reactor types.
- **Moderator:** Graphite or water slows neutrons to thermal speeds so they are more likely to cause fission (U-235 has a much higher fission cross-section for slow neutrons).
- **Control rods:** Made of boron or cadmium, which absorb neutrons. Inserting rods reduces the neutron population, slowing the reaction. Withdrawing rods increases the reaction rate. This maintains criticality (exactly one new fission per fission).
- **Cooling fluid:** Water or liquid sodium circulates through the reactor core, removing the heat generated by fission. The heat is used to raise steam, which drives turbines to generate electricity.
- **Safety:** Emergency shutdown systems insert all control rods if the reactor overheats or malfunctions.

Advantages and Disadvantages of Nuclear Fission

Advantages	Disadvantages
Very high energy output per fission (~200 MeV)	Radioactive waste produced (long-lived fission products, actinides)
No CO_2 emissions (zero-carbon energy)	Long-term waste storage problem (1000s of years of containment required)
High power density (small reactor, large output)	Risk of catastrophic accidents (meltdown)
Reliable (not weather-dependent like wind/solar)	High decommissioning costs
	Weapons proliferation risk (enriched uranium)

4.4.10 Nuclear Fusion (Triple Science)

Nuclear fusion: the combining of two light nuclei to form a heavier nucleus, with release of energy. Fusion requires very high temperature and pressure to overcome the electrostatic

> repulsion between the two positively charged nuclei.

Fusion Reaction Example

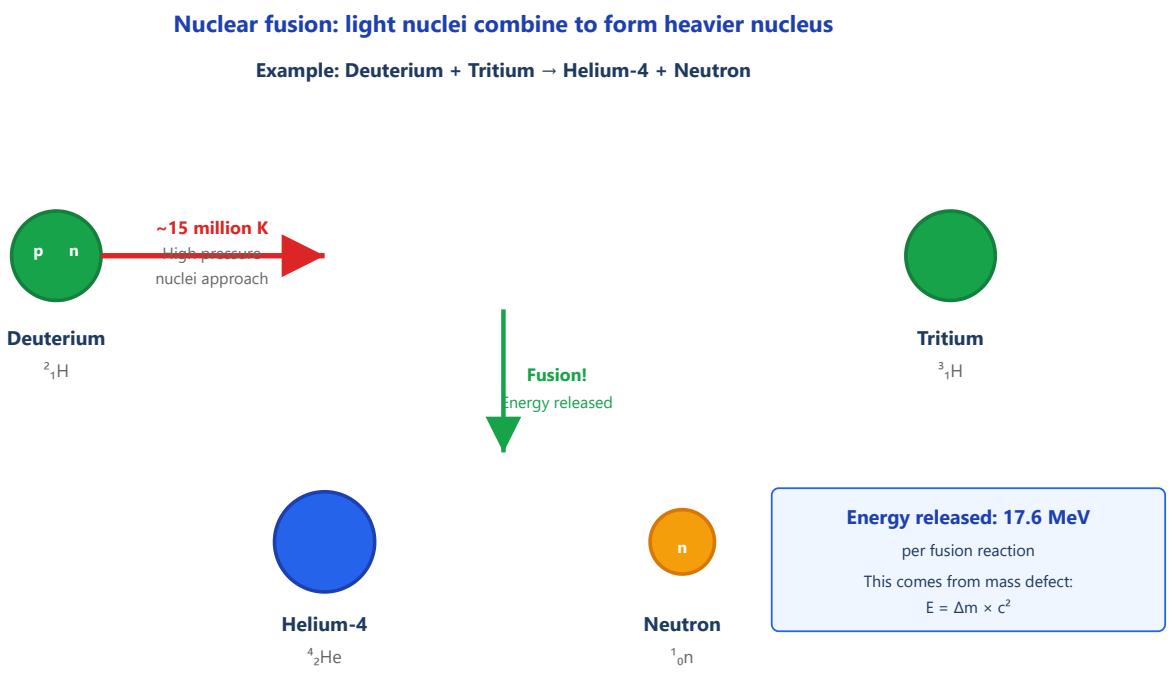
```
Hydrogen isotope fusion:
²₁H + ³₁H → ⁴₂He + ¹₀n + 17.6 MeV

(Deuterium + Tritium → Helium-4 + Neutron)

This is the reaction proposed for ITER (International Thermonuclear Experimental Reactor
```

Fusion in Stars

The core temperature of the Sun is ~15 million K. Hydrogen nuclei (protons) fuse to form helium in the proton-proton chain:

```
Simplified: 4¹₁H → ⁴₂He + 2⁰₊₁e (positrons) + 2νₑ (neutrinos) + ~26.7 MeV
```

This reaction has powered the Sun for ~4.6 billion years and will continue for another ~5 billion years.



Why Fusion Releases Energy: Mass-Energy Equivalence

Einstein's famous equation: `E = mc²`

In a fusion reaction, the total mass of the products is slightly less than the total mass of the reactants. This "missing" mass is converted into energy according to $E = mc^2$. Even a tiny mass difference produces enormous energy because c is so large (3×10^8 m/s).

Example: In D-T fusion (deuterium + tritium), ~3.5 MeV comes from the mass defect. The rest of the 17.6 MeV comes from the kinetic energy of the nuclei and the release of binding energy stored in the nuclear bonds.

Fusion vs Fission

Property	Fission	Fusion
Process	Heavy nucleus splits	Light nuclei combine
Temperature required	Room temperature (neutron trigger)	~100 million K
Control	Can be controlled (rods, moderator)	Very difficult to control; no commercial reactors yet
Radioactive waste	Long-lived fission products (1000s of years)	Minimal waste (neutron activation of reactor walls)
Fuel availability	Uranium-235 is limited (~100 years at current rates)	Deuterium and tritium from seawater; abundant
Current use	~430 commercial power reactors worldwide	Only in stars and thermonuclear weapons; experimental research (ITER)

Challenges in Achieving Fusion Power

- **Temperature:** Nuclei must be heated to ~100 million K to overcome electrostatic repulsion. This is roughly 7× the core temperature of the Sun.

- **Confinement:** At such high temperatures, matter becomes a plasma (ionised gas). The plasma must be confined using magnetic fields (tokamaks) or laser compression (inertial confinement fusion).
- **Energy gain:** The energy needed to heat and confine the plasma must be less than the energy released by fusion. This is the "break-even" condition; no one has achieved this yet (though ITER aims to).
- **Sustained reaction:** A fusion power plant must run continuously, not just produce one pulse.

> **Exam Tip:** Common exam question: "Compare nuclear fission and fusion." Key differences: Fission splits heavy nuclei (easy to trigger, but produces long-lived waste). Fusion combines light nuclei (produces little waste, but requires extreme conditions). Both release energy from mass defect ($E = mc^2$). Fusion powers the Sun; fission powers nuclear power plants.

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